

I — Contributions to research and development

a. Articles published or accepted in peer-reviewed journals

Rouleau-Desrochers, C., Van der Meersch, V., Buonaiuto, D.M., Chamberlain, C.J., Loughnan, D., Pederson, N., & Wolkovich, E.M. (2026) Warmer and longer growing seasons increase growth for juvenile trees in a common garden. *Global Change Biology*. Submitted (MSc work).

Wolkovich, E.M., **Rouleau-Desrochers, C.**, Garcia de Cortazar-Atauri, I., Walker, M.A., Lacombe, T. (2025) Building a more predictive model of Terroir for the Anthropocene. *Plants, People, Planet*. DOI: 10.1002/ppp3.70055 (BSc work).

b. Other peer-reviewed contributions

Rouleau-Desrochers, C., Van der Meersch, V., Pederson, N., & Wolkovich, E.M. (2026) Evidence that growing season length and tree growth are decoupled in an urban arboretum. *Canadian Society for Ecology and Evolution (CSEE)*. Full oral presentation at a national-level conference (MSc work).

Rouleau-Desrochers, C., Pederson, N., & Wolkovich, E.M. (2025) How growing season length affects tree growth in a common garden study. *EcoEvo*. Oral presentation at a regional-level conference (MSc work).

c. Non-peer-reviewed contributions

Rouleau-Desrochers, C., Patry, É., Gingras, G., Ucler, J., Vézina-Doré, M., Paillé, O., Moghaddam, Y.Y. (2024) “Un été parti en fumée.” *Le Point Bio*.

II — Most significant contributions to research and development

1. Warmer and longer growing seasons increase growth for juvenile trees in a common garden (Global Change Biology and CSEE 2026). In this, I address a long-standing contradiction in phenology research: while extended growing seasons are widely assumed to increase tree growth, recent work has repeatedly failed to find this link, largely because it focused on early-season leafout, when cool temperatures limit photosynthesis. I designed and led an analysis testing how juvenile tree growth in four Betulaceae species responds to both season length and thermal sum, using start- and end-of-season phenology data across four provenances. I found that warmer seasons were a better predictor for growth than season length in most species, and that end-of-season phenology was a stronger predictor than start-of-season, challenging assumptions about when in the season growth actually happens. I also tested for provenance and found little evidence of local adaptation, suggesting these responses may generalize across species ranges. We targeted *Global Change Biology* given its readership of ecologists and modelers working on climate-driven shifts in ecosystem processes. My work shows that season-length effects on tree growth are species- and timing-dependent, with direct implications for how forest regeneration and range-expansion models incorporate phenology into large-scale carbon models.

2. Predictive model of Terroir under climate change (Plants, People, Planet, 2025). This paper tackles an urgent applied problem: current vine-planting choices and limited genotype $\times$ environment data undermine adaptation planning under climate change. I synthesized fragmented genotype $\times$ environment datasets into a proposed meta-analysis framework and designed a global data-gap map that reframed the paper’s core argument: predictive models must explicitly account for missing data to guide adaptation. I contributed to shaping the manuscript’s conceptual structure, ensuring the analyses were accessible to both

ecologists and practitioners. We chose Plants, People, Planet for its mandate to bridge fundamental plant science and applied practitioner audiences, matching our goal of making the findings usable for viticulture planning. My work provided the methodological backbone that made the study's recommendations actionable under climate change.

3. Public communication on Québec wildfires (Le Point Bio, 2024). I wrote and co-produced a public-facing article explaining the causes and health impacts of the 2023 Québec wildfires shortly after the events. Drawing on my boreal-forest ecology background, I led expert interviews and framed complex findings in accessible language. This work reflects my commitment to turning research into public understanding, and helped me practice rapid, accurate science communication in a high-stakes context.

III — Applicant's statement

Research experience: The first time I really fell into natural-science research was in 2021 with Dr. Pierre Drapeau during which I travelled across Québec investigating woodpecker damage to wooden electrical infrastructures—a project that taught me rigorous field protocols, database management and the basics of scientific coding. In 2022, I held my first Undergraduate Student Research Award (USRA) and carried out intensive fieldwork on Pileated woodpecker habitat in remote boreal forest: long days of banding, nest-searching and tree-classification built my field skillset. Alongside this, I leveraged 20 years of nesting data to understand how Pileated woodpecker nesting phenology is affected by climate change—my first independent statistical project. A second USRA at UBC in 2023 (supervised by Drs. Frederik Baumgarten and E.M. Wolkovich) exposed me to large-scale experiments. During the summer, I designed and implemented a defoliation experiment to test tree resilience, maintained a 1,000-tree trial, deployed dendrometers and analyzed growth responses to treatments.

In 2024, I conceived and led a greenhouse experiment (foundation of my first MSc chapter), where I formulated hypotheses on plant-climate interactions, designed treatments to test them, and integrated climate and phenological data into statistical models. Alongside this, I also joined field campaigns in Smithers (BC) and Mount Rainier (WA) to study forest regeneration dynamics, where I improved my forestry methods and learned the basics of dendrochronology. In 2025, while continuing my greenhouse experiment I led the fieldwork at Smithers and visited the Arnold Arboretum of Harvard University to collect tree cores, deepening my expertise in dendrochronological methods.

Relevant activities and distinctions:

Prior to my current academic path, I earned degrees in the restaurant and wine industries and spent nearly ten years working in the field, where I built a strong work ethic and sense of rigour. The transition from the food and beverage industry to academia happened while working for six months in a vineyard in New Zealand; I realized that my ties to nature were stronger than I expected. Thereafter, I enrolled in a Bachelor's degree in biology, majoring in ecology. In 2025, during the first year of my MSc degree, I deepened my quantitative skills through a biomathematics class, a three-week workshop on hierarchical Bayesian models, and the Statistical Rethinking course, presenting and refining my models in a biweekly Bayesian working group where peers comment and suggest improvements.

My dedication and constant efforts during my undergraduate studies at Université du Québec à Montréal were recognized with the Governor General's Academic Silver Medal, Dean's List of honours, 13 competitive scholarships totalling \$76,750, and a perfect GPA of 4.30/4.30. For my MSc work, I received three competitive scholarships totalling \$28,750, an honourable mention for my EcoEvo talk and the 'Honours' mention for my thesis work. I have maintained active outreach throughout: I was interviewed at the 2023 UQÀM scholarship ceremony, featured on Biochambers for my growth-chamber work, hosted an open house at the Canadian Society for Ecology and Evolution (2024) and presented my work twice to the Treepotters community scientists at Harvard (2025-2026).

Across the two years leading my experiment, I developed strong leadership qualities as I trained and supervised 15 undergraduate students in growth-chamber protocols, phenological monitoring and greenhouse management, two of whom are currently pursuing graduate studies. I also led the team for Smithers field-work in 2025. In addition to my mentor role in my laboratory, I was a teaching assistant for FRST303 (2024-2025) and FRST304 (2025-2026), during which I ran office hours and gave supplementary materials to help students grasp ecological and plant physiology concepts. Outside my lab work, since 2021, I have mentored twelve international students during their first months in Canada, tutored seven peers in biochemistry and statistics, and volunteered with BUDR to advise eight students on applying to graduate school.

My wildlife photography work complements my outreach: I won the 2026 Biodiversity Research Center (UBC) photo competition, and my photos have appeared on the cover of *Le Point Bio* (2024), in an article of the Proceedings of the National Academy of Sciences (PNAS), and in *Le Rappel*.